United International University

School of Science and Engineering

Mid-term Examination; Year 2026; Trimester: Spring
Course: BIO 3105/3107; Title: Biology for Engineers/Biology
Section: All
Full Marks: 30; Time: 1 hour 30 minutes

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*There are 5 Questions: 1, 2, and 3 are mandatory, and you must answer **either** 4 or 5 (any one)*

1.	a. What are the levels of organization of a protein structure?	1	CO1
	b. In how many ways, genetic variation can occur during meiosis? Briefly explain each type with necessary figure.	4	CO2
	c. Which cell organelle is known as the 'packaging center', and how are materials released from the cell by exocytosis?	3	CO1
2.	a. Outline the process by which DNA instructions lead to the formation of a protein.	2	CO3
	b. What are the key principles of modern cell theory?	3	CO1
	c. How does the prophase stage of mitosis differ from prophase I of meiosis?	3	CO2
3.	a. How can you leverage your knowledge of CSE to solve a biological problem? To answer this question, you must define a problem related to biological science and you should implement the concept(s) of CSE to solve that problem.	4	CO2
	b. What is oncogene? How do they cause cancer?	2	CO3
	c. What are the basic characteristics of genetic codon?	2	CO1
4.	a. How can Cryo-EM technique be used to determine 3D structures of biological macromolecules?	3	CO4
	b. After running genetic tests, it was observed that a child has three copies of chromosome 21. What is the name of this condition? Which technique would you utilize to obtain this observation?	3	CO4
5.	a. Explain**,** with the help of a labeled diagram, the stages of the cell cycle and indicate the major checkpoints where regulation occurs.	3	CO1
	b. Solutes are transported across membrane via active or passive transporters. When might we need to use active transporter to transport solute across membrane? Justify.	3	CO3

CO1: Describe different biological quantities.
CO2: Apply the knowledge of biological systems in a real-life problem.
CO3: Design several biological systems with constraints.
CO4: Explain several procedures for solving biological systems within constraints.